

Climatic drivers and biological constraints drive masting dynamics

Victor, Mike, Lizzie

May 10, 2026

Abstract

Masting—the synchronous ‘boom-and-bust’ reproduction observed across many tree species—shapes forest regeneration globally. Decades of research have identified climate as a potential driver of masting, raising concerns climate anthropogenic climate change may disrupt it, but cannot well predict synchrony. Here, we integrate a well-known constraint on reproduction—competition between floral buds and fruit maturation that drives a trade-off between the two processes across years—that operates at the individual tree level with known climatic drivers on masting in the temperate zone to forecast masting with climate change. Using long-term seed production data from individual beech trees across England, we scaled up tree-level predictions to population-level dynamics, and found strong evidence for synchrony, with 87% of trees within a population sharing the same reproductive state in any given year since 1980. Synchrony between populations did not appear to have declined significantly over the last decade despite significant warming (84% before 2015 to 83% after). Over this period, cold summers were the main controller of synchrony because they drove most trees into a shared low reproductive state, providing the conditions for synchronous reproduction when a warm summer followed. Warm summers increased the probability that trees transitioned into a high reproductive state, but individual-level reproductive constraints prevented runaway increases in masting frequency, even when forecasted under extreme warming scenarios. While our results suggest the loss of cold summers could reduce synchrony in the future, the current interplay of individual-level constraints and population-level climatic cues suggests past and continued synchrony.

Introduction

The acceleration of climate change may cause abrupt ecological effects worldwide (Trisos *et al.*, 2020). Increasing extreme conditions could disrupt key ecological processes, and drive ecosystems toward critical transitions (Wernberg *et al.*, 2016). Across biomes, recent work has suggested major shifts in pests and pathogens, while the local extirpation of rare species and increase in invaders across taxa has driven biotic homogenization (Carlson *et al.*, 2025; Dornelas *et al.*, 2014; Daru *et al.*, 2021). Forests in particular appear to be shifting, with lagging elevational and latitudinal range expansions (CITE) alongside increases in adult tree mortality. This has driven growing concerns about the future of forest carbon sinks globally (Albrich *et al.*, 2020; Forzieri *et al.*, 2022; McDowell *et al.*, 2020) and growing interest into the mechanisms that underlie forest regeneration (Foest *et al.*, 2024; Hacket-Pain *et al.*, 2025; Chakraborty *et al.*, 2024).

Regeneration in many temperate and tropical forests depends on the widely observed phenomenon of masting. Masting—defined by ‘boom and bust’ cycles of reproduction that are synchronous across most individuals of a population (Janzen, 1978)—occurs across many tree species and is hypothesized to drive fitness benefits during reproduction and the early stages of establishment (Kelly, 1994; Bogdziewicz *et al.*, 2024). The most commonly invoked fitness benefit, ‘predator satiation,’ hypothesizes that years of high seed production overwhelm seed predators and thus allows a higher proportion of seeds and seedlings to escape predation and establish (Janzen, 1971; Kelly, 1994). Fitness benefits of masting may also operate earlier in reproduction, by increasing pollen exchange and genetic outcrossing (Carlson *et al.*, 2014; Bontrager & Angert, 2019). These different hypotheses vary in their exact mechanism and life stage. Yet, they all imply that the benefits of masting depend on an economy of scale where most individual trees within a population (or location) must be synchronous.

Despite the importance of synchrony to yielding fitness benefits, how synchronous masting is and how it occurs remain poorly understood. Most hypotheses for how synchrony emerges focus on shared environmental cues at the species-level (Kelly & Sork, 2002; Bogdziewicz *et al.*, 2024). Among many potential cues, an abundance of work suggests that ideal climatic conditions during bud formation may lead to masting (Nussbaumer *et al.*, 2018; Journé *et al.*, 2025). In temperate systems, ideal conditions include sufficiently warm temperatures during the growing season before masting—when reproductive buds generally develop—alongside mild spring conditions that do not cause tissue loss after buds form (i.e., no late spring frosts, Vacchiano *et al.*, 2017; Gallego Zamorano *et al.*, 2016). Climate change, however, is rapidly shifting temperatures across the spring and summer, including the prevalence of spring frosts (Zohner *et al.*, 2020; Chamberlain *et al.*, 2021), and thus could disrupt masting dynamics. Such changes would have major implications for forest resilience (Bogdziewicz *et al.*, 2023; Foest *et al.*, 2024), but forecasting them requires better understanding how these climatic cues affect reproductive biology, from buds to fruits.

Most temperate woody species develop reproductive buds through a multi-year process. In the common two-year reproductive cycle, bud formation begins the year before flowering and seedset—potentially explaining why summer temperatures from the previous year appear to trigger masting (Geber *et al.*, 1997; Wilkie *et al.*, 2008). The following year some of these buds may be lost to spring frost—potentially reducing the number of seeds in what otherwise have been a masting year. This two year cycle, however, also introduces a constraint in most species. Because floral buds for the following year develop simultaneously with current-year fruit (Geber *et al.*, 1997), a large fruit crop year can depress bud initiation (through hormonal inhibition and within-plant resource competition Monselise & Goldschmidt, 1982; Bangerth, 2009; Milyaev *et al.*, 2022). The result is physiological constraints on flower and fruit development that is often invoked to explain why many woody species show alternate bearing—with a large crop production year (‘on-year’) often followed by one or several years of low production (‘off-years’, Monselise & Goldschmidt, 1982; Wilkie *et al.*, 2008).

These constraints may limit how climate—and climate change—can affect seed production. Ideal climatic conditions may trigger higher investment in floral buds only for individual trees not

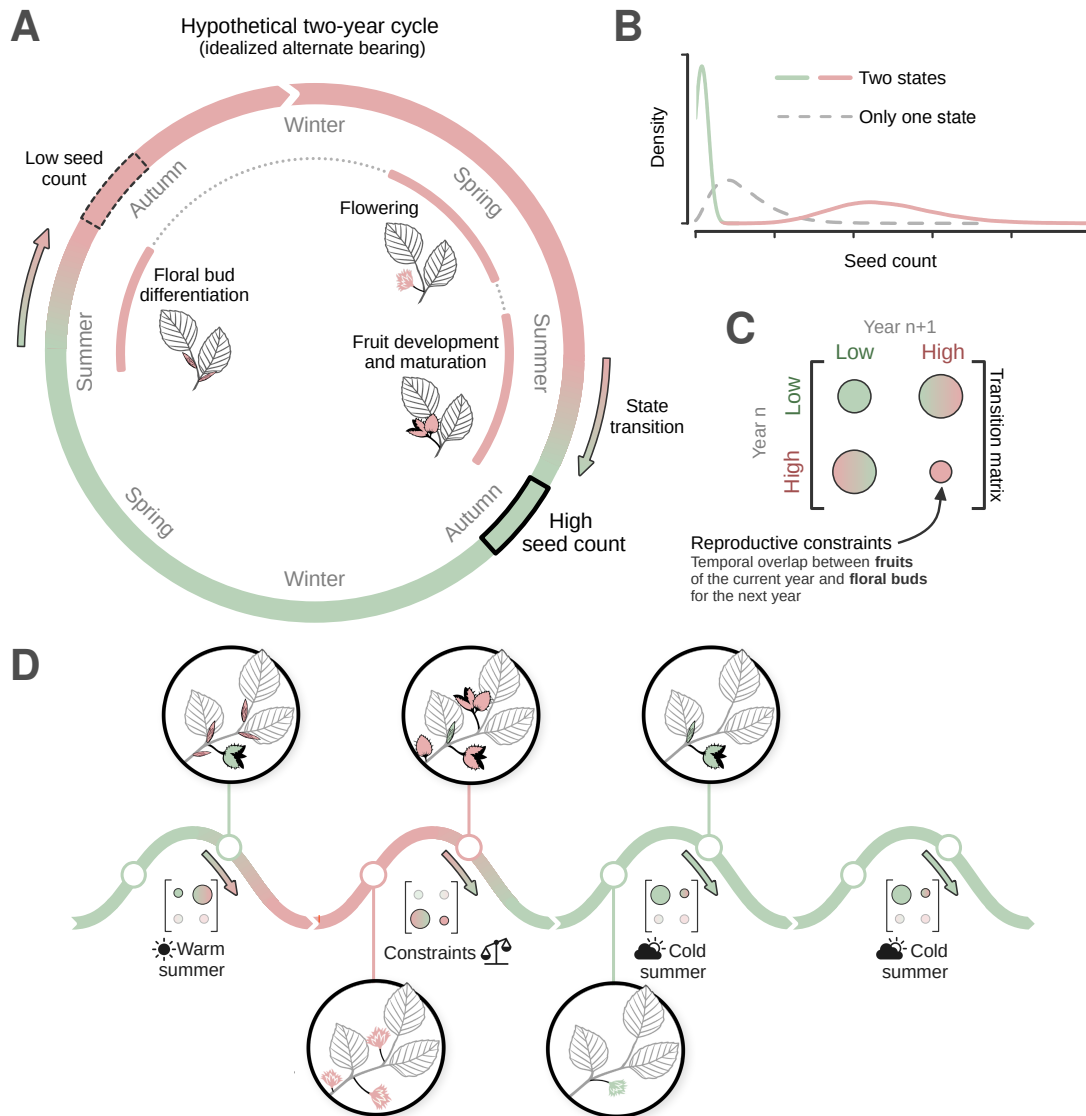


Figure 1: **Conceptual figure.**

already in a high reproductive state. Warm summers would therefore not affect all trees equally—trees in a high reproductive state would be unaffected, while those in a low reproductive state would likely increase their probability of transitioning to a high reproductive state. Once a reproductive state has been established, then spring frost may have an effect, with damage from frosts most likely to reduce potential seed production for those trees in a high reproductive state. These individual-level reproductive dynamics may then scale up to generate population-level synchrony, with masting emerging through a combination of climatic variability and reproductive constraints. Accounting for these differential effects of climate could help forecast how climate change may alter masting as an emergent phenomenon of individual reproductive cycles.

Here we test how climate affects masting by explicitly modeling how climatic cues interact with individual reproductive constraints to generate population-level synchrony. Using seed data collected on individual trees across populations of beech forests of England since 1980, we first test whether individual trees show evidence for alternative reproductive states that could lead to masting at the population level. We then use these states to examine how synchrony arises from the combination of reproductive constraints, at the individual-level, and climatic cues that operate over larger scales. Finally, we leverage these results to forecast how future warming may affect both individual tree-level seed production and population-level synchrony.

Results and discussion

Following the paradigm that masting is defined by of ‘boom and bust’ reproduction, our model identified high and low reproductive years as alternative latent states (Fig. 2A-B). In high (‘boom’) reproductive years, individual trees produced 168 seeds on average (90% uncertainty range: 163 – 173), while in low (‘bust’) reproductive years they produced 52 seeds on average (48 – 56 seeds per tree). Seed number per tree in low and high reproductive years, however, overlapped significantly (Fig. 2B). The difference between these two states thus occurs mostly in the opposite extremes—years in the high reproduction latent state included very high seed counts, which would almost never be possible in the low reproduction state, while years in the low reproduction state were characterized often by extremely low counts (between 0 and 10 seeds) almost never seen in the high reproductive state (Fig. 2B). Across populations, years of high seed production—mast years—were almost always separated by periods of low seed production (Fig. 2C).

Transitioning between reproductive states (high or low) depended on a tree’s state in the previous year (Fig. 2D). In a year with average climatic conditions (summer temperatures around 19.7°C), a tree in a low reproductive state was most likely to transition to a high reproductive state the next year, with a probability of 64% (61 – 68%), and a 1.8 times lower probability (36%, 32 – 39%) of staying in a low reproductive state (Fig. 2D). Under the same average climate, a tree in a high reproductive state was almost certain to transition to a low reproductive state, with a probability of 99% (97 – 100%), and very unlikely to remain in the same state (1%, 0 – 3%). These results suggest that trees on average shift into a new reproductive state most years, but are more likely to remain in a low reproductive state, supporting the idea of non-mast years being more

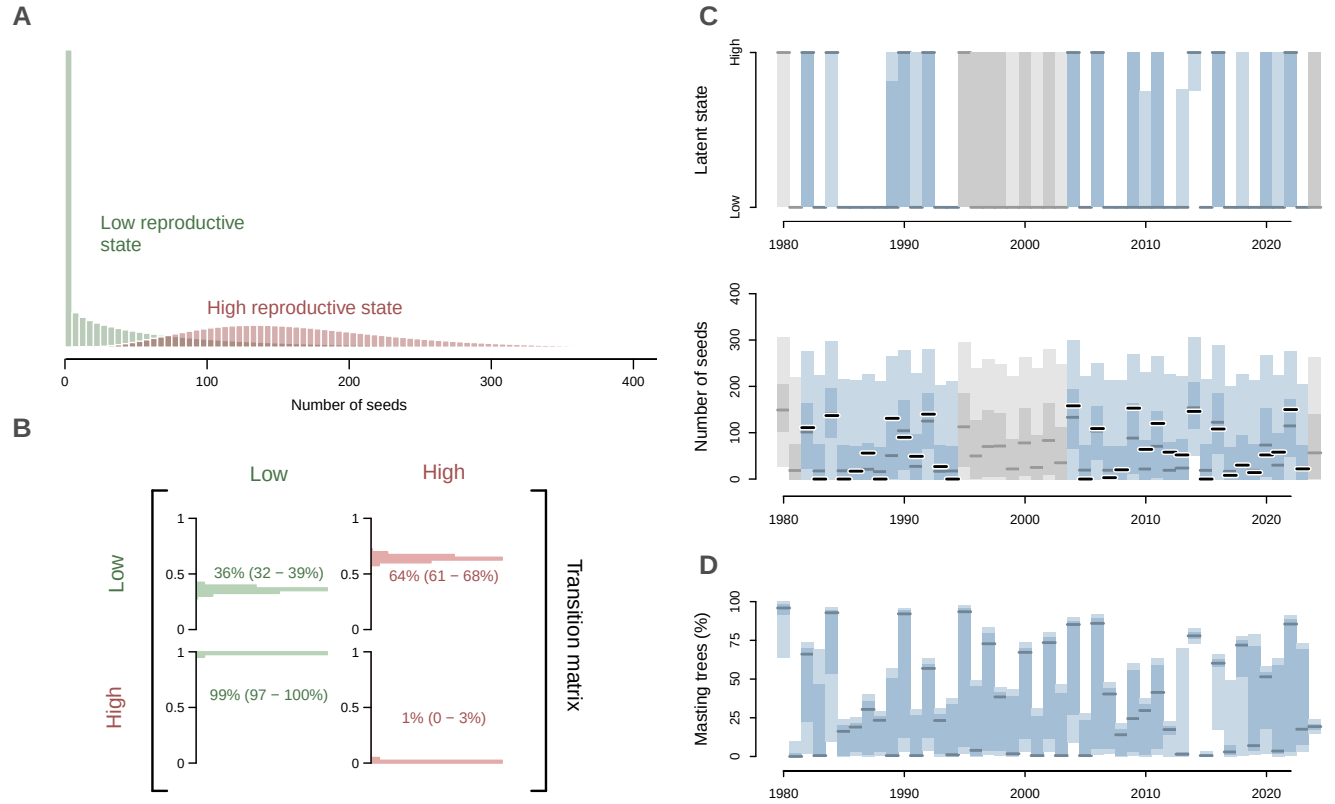


Figure 2: Distinct tree reproductive states generates variability at the population level. **(A)** The model identifies two distinct reproductive latent states that correspond to two different level of seed production. **(B)** The differences in transition probabilities between the two latent states reflect endogenous constraints that shape tree-level reproductive cycles. **(C)** At the tree level (illustrated here for one random tree), the alternation between these two latent states across time (upper plot) generates low and high seed production years (lower plot). **(D)** When added together across all populations, these individual cycles generate variability and synchrony in seed production across years. In **(C)** and **(D)**, the blue line represents the median model prediction, while the shaded areas represent the 25–75% and 5–95% quantile ranges. The black lines represent the observed seed counts, and the grey lines and areas indicate model predictions for years when observations were missing.

113 frequent than mast years (Kelly, 1994).

114 Climate strongly affected these transition dynamics. Similar to previous studies (Nussbaumer
115 *et al.*, 2018; Vacchiano *et al.*, 2017; Journé *et al.*, 2024), we found that warm summer temperatures
116 increased the probability that trees in a low-reproduction state transitioned to a high-reproduction
117 state (Fig. 3A). For example, the transition probability during a warm summer at 23°C was 8.1
118 times higher than during a 17°C summer (98% vs. 12% respectively, and ranges of 96 – 99% vs.
119 9 – 16%). Additionally, our results support the common hypothesis that spring frosts may limit
120 seed production (Augspurger, 2009), with seed production for trees in a high reproductive state
121 reduced by -20.1% (-26.1 – -13.9%, Fig. 3B) given a three-fold increase in a metric of frost risk
122 (growing-degree days until last frost, Vitasse & Rebetez, 2018), but no evidence that warm springs
123 affected seed production (Supp. Fig.). Given the reproductive constraints that prevent beeches
124 and many other woody plants from repetitive years of high fruit (from the competitive overlap of
125 flower bud formation and fruit development, Monselise & Goldschmidt, 1982; Geber *et al.*, 1997),
126 we assumed the transition from a high to a low reproductive state was not affected by climate (we
127 found strong support for this when we relaxed this assumption, Supp. Fig.).

128 Individual tree-level reproductive states (high or low) scaled up to produce population-level
129 seed production that was synchronous, both within and between populations (sites). Within pop-
130 ulations, 87% (84 – 89%) of trees shared the same reproductive state on average each year, while
131 synchrony across populations was only slightly lower (the proportion of trees in the same state
132 across all populations was 84%, 81 – 86%). This high level of synchrony across populations could
133 suggest that synchrony occurs across wide spatial scales, as sometimes suggested (Bogdziewicz *et al.*,
134 2021a; Journé *et al.*, 2024), but given that these populations were not widely separated spatially
135 (Supp. Mat.), these results only provide evidence for synchrony over smaller spatial scales, as often
136 suggested (Koenig & Knops, 1998; LaMontagne *et al.*, 2020). Whether this scale is sufficient for
137 fitness benefits depends on the exact mechanism (e.g., predator satiation or pollination efficiency)
138 and the species-level attributes (e.g. predators foraging ranges or wind dispersal distances, which
139 depend on seed shape, size and the local landscape Davies & MacPherson, 2024). For example, if
140 beech masting in England mainly serves to overwhelm seed predators, such as wood mouse or bank
141 vole (Sachser *et al.*, 2025), then masting synchrony at the population level (given that sites are on
142 average 224 km apart, and the two closest sites 1 km apart) should be sufficient to maintain the
143 regeneration benefits of masting.

144 Climatic cues—especially cold summers—appeared critical to synchronizing the reproductive
145 cycle of individual trees to produce the population-level synchrony that often defines masting (Kelly
146 & Sork, 2002). While 36% of trees in a low reproductive state are likely to stay in that state, cold
147 summer temperatures prevent an additional subset from transitioning to a high reproductive state
148 (because cold summer temperatures reduce the probability of transition to a high reproductive
149 state). Thus natural reproductive constraints combined with cold summers can leave most trees
150 in a low reproductive state (Fig. 3A). This synchrony of low reproduction appears to then drive
151 synchrony of high reproduction when a cold summer is followed by a warm summer; in such years
152 most trees responded consistently to the warm temperatures that promote floral induction, and

transitioned together into a high reproductive state (Fig. 3C), thus generating a mast year at the population scale. This interplay between individual-level constraints and population-level climatic cues provides a physiological mechanism for reproductive synchrony that can persist over multiple years, and offers a simple explanation for previous findings that masting depends on the temperature difference between successive summers (the ΔT model; Kelly *et al.*, 2012).

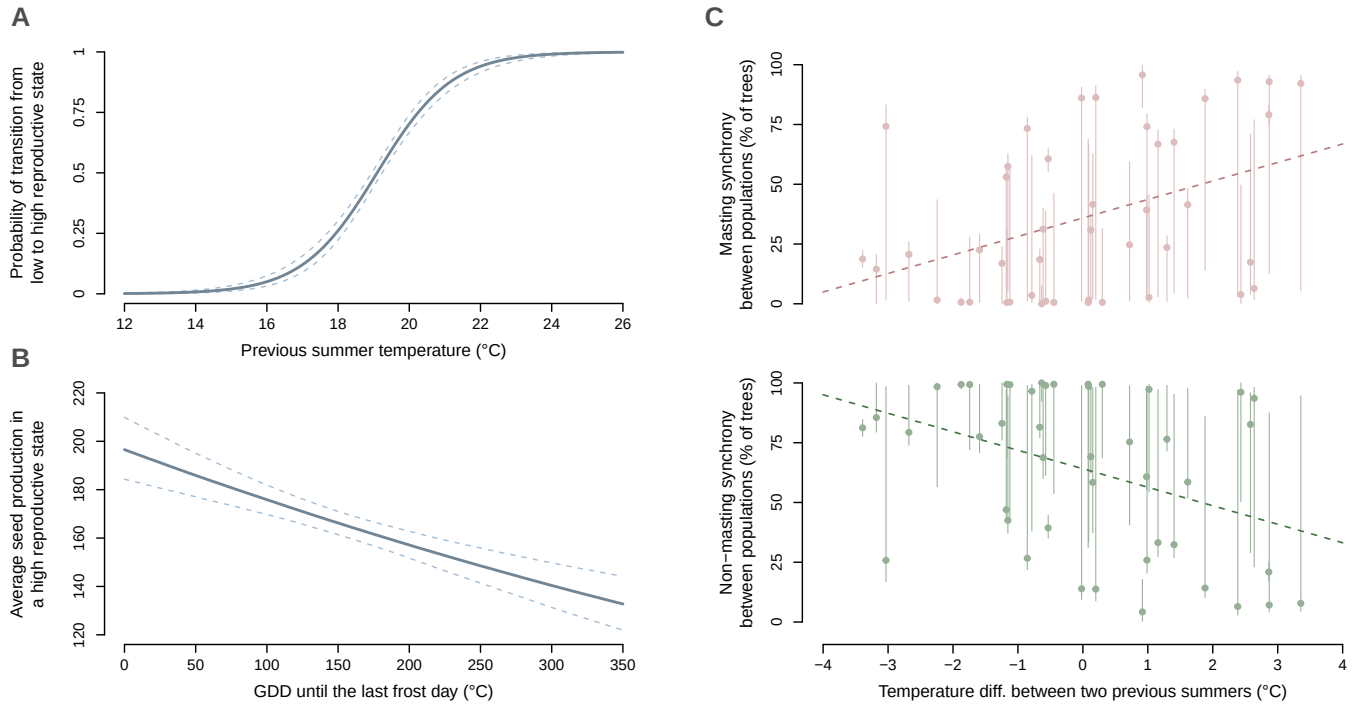


Figure 3: Climate conditions impact both transition probabilities between states and seed production in a high reproduction state. (A) Average maximum temperature in July and August of the previous year increases the probability of transition from a low-reproduction to a high-reproduction state. **(B)** The accumulated growing-degree days (GDD) until the last frost day decreases the number of seeds produced in a high-reproduction state. Plants that have accumulated more GDD before a frost event are phenologically more advanced and thus face a higher risk of frost injury (Vitasse & Rebetez, 2018). **(C)** The interactions of previous summer conditions and previous latent states explain the apparent two-year lag effect of climate on observed synchrony across populations. In **(A)** and **(B)**, the solid red line and dashed lines respectively represent the median and the 5% and 95% quantiles, while the gray boxplots show observed conditions (1980–2022).

Our results appear to support the hypothesis that increasing summer temperatures with climate change could disrupt masting (Bogdziewicz *et al.*, 2021b; Foest *et al.*, 2024). This hypothesis—previously called breakdown—predicts that warm summers will drive trees to be in high reproductive state more frequently and thus years of low reproduction will become rarer, potentially leading to lower individual tree growth (Koenig & Knops, 1998; Hacket-Pain *et al.*, 2025) and reducing variability and synchrony in seed production at the population level. While we do find that warmer summers increase the probability of trees transitioning into a high reproductive state—and thus increasing summer temperatures would increase the frequency of high seed production years—our

166 results suggest that reproductive constraints prevent major shifts in the frequency of high seed
 167 production years. Indeed, we found even with extreme summer warming (7°C in one century, cor-
 168 responding to the warmest regional climate model projections, Schumacher *et al.*, 2024), only 52%
 169 (40 – 64%) of trees in a population would mast in any given year (Fig. 4). This result remains
 170 even when we relax reproductive constraints and allow warm summers to increase the probability of
 171 trees staying a high-reproductive state (Supp. Fig.). Even with this additional effect, trees cannot
 172 become stuck indefinitely in a highly reproductive state. On the contrary, we found a ‘breakdown’
 173 because of climate change would require summer warming to have a similar impact on both the
 174 transition and persistence into a high-reproduction year—an hypothesis not supported by the data
 175 we used here (Fig. 4).

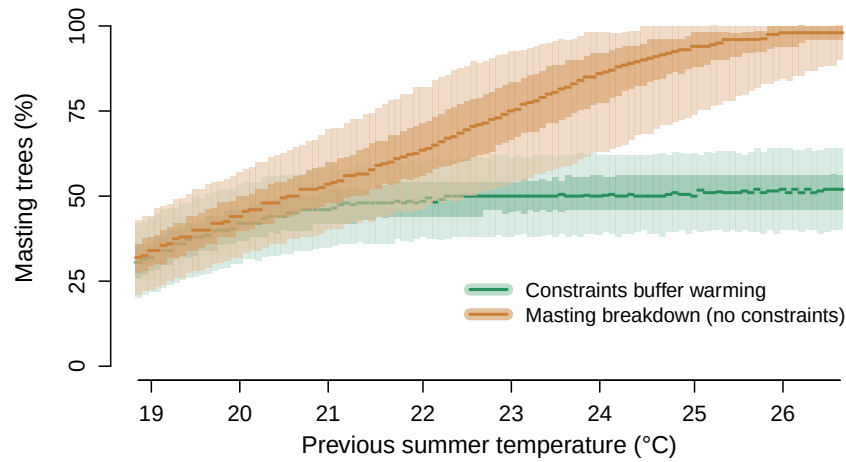


Figure 4: **Endogeneous constraints prevent masting breakdown under a significant summer warming.** We predicted the percentage of masting trees for 50 new trees in an average population in England under a strong summer warming of 7°C in one century—which corresponds to the warmest regional climate model projections (Schumacher *et al.*, 2024). The green line and shaded areas (median, 25–75% and 5–95% quantile ranges) represent the predictions with the model fitted on the data used in this paper. With this model, the fitted transition probabilities indicate that trees cannot remain continuously in a high-reproduction state. The orange line and shaded areas (median, 25–75% and 5–95% quantile ranges) represent the predictions from an hypothetical model where summer warming increases the probability of persisting into a high-reproduction state—and would lead to most trees masting every year (i.e. reproductive breakdown). These predictions are not supported by the data used here.

176 These results highlight how biological constraints on plant reproduction may limit runaway
 177 effects of climate change. For masting, floral bud initiation at the individual tree level is promoted
 178 by warm summer temperatures, potentially leading to a higher number of flowers and fruits in the
 179 following year (Vacchiano *et al.*, 2017; Bogdziewicz *et al.*, 2024; Journé *et al.*, 2024). In the next
 180 summer, however, the development of a larger fruit load inherently imposes a trade-off that limits
 181 the tree from initiating a large number of floral buds again (Monselise & Goldschmidt, 1982; Wilkie
 182 *et al.*, 2008). These individual-level constraints scale up to the population level by preventing the
 183 entire population from producing large seed crops every year—and thus also preventing an unlimited

184 amplification of climate effects on masting. Our results highlight that constraints at the individual
 185 level are critical for accurate forecasts of population-scale dynamics (CITE?). Further this approach
 186 can help identify the spatial scale at which changes occur, including possible shifts in synchrony.

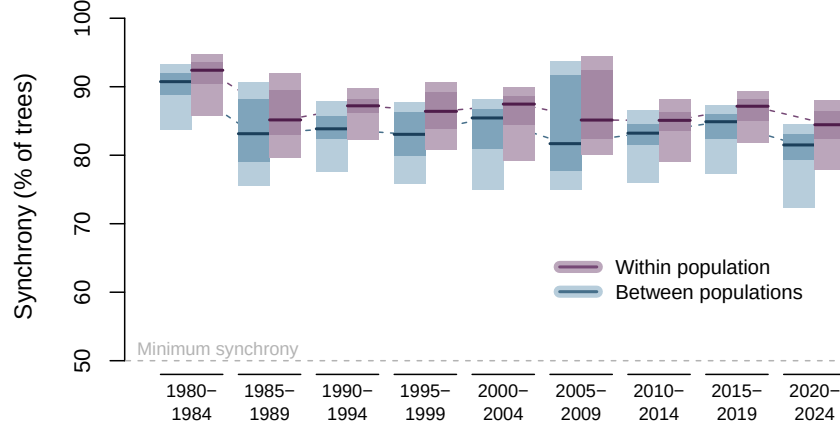


Figure 5: **Reproductive synchrony arises from populations of individual trees.** Blue line and shaded areas (median, 25–75% and 5–95% quantile ranges) represent the synchrony within population, i.e. the average proportion of trees of the same population that are in the same state. Some populations can be in a non-masting year, while some population can be a masting year. Purple line and shaded areas (median, 25–75% and 5–95% quantile ranges) represent the synchrony across populations, i.e. the proportion of trees that are in the same state across all populations. Since our model includes only two states, the minimum synchrony is 0.5 (i.e. at least 50% of the trees are in the same state).

187 In contrast to previous results we found no clear evidence of declines in synchrony in recent
 188 decades, during a period of significant warming (Bogdziewicz *et al.*, 2020; Foest *et al.*, 2024).
 189 Synchrony across trees within a population varied over time but has not clearly changed in recent
 190 periods (recent declines in 2015 onward are within the 90% range of previous periods, Fig. 5A).
 191 Similarly, synchrony between populations has been quite stable, from a previous mean of 84.3%
 192 (81.1 – 87.2%) to one of 82.8% (78.9 – 85.2%) since 2015. This stability in synchrony across both
 193 scales suggests that masting in these beech populations may be more resilient to warming than
 194 previously predicted (Bogdziewicz *et al.*, 2021b).

195 Whether masting can continue to support forest regeneration in the next decades, however, may
 196 depend on how synchrony will respond as climate continues to warm. While we find no evidence for a
 197 positive feedback of summer warming that would trigger runaway seed production at the individual
 198 level, our results predict possible changes at the population level in synchrony as we lose the cold
 199 summers. Because cold summers act as potential hinge points for population synchronization,
 200 several consecutive years without cold summers could progressively decrease synchrony within a
 201 population (Fig. 6). Our predictions show how repeated warm summers could lead to low synchrony
 202 (52% in Fig. 4) not because they drive high seed production from most trees getting stuck in a
 203 high reproductive state (as previously suggested, Bogdziewicz *et al.*, 2020; Foest *et al.*, 2024), but
 204 because of the loss of cold summers that drive most trees in a population into the same reproductive

states. Determining the extent to which this will happen in the coming decades, and producing robust spatiotemporal forecasts, will require better measurements of the temporal and spatial scales at which synchrony appears and disappears across years (LaMontagne *et al.*, 2020). Our results suggest models of masting—and its inherent synchrony—should build from individual-level seed production data, which is still incredibly rare. Teasing out the effects of warm summers versus spring frosts will likely also require temporally finer resolution data than seed counts—including flowering Nussbaumer *et al.* (2020) but also ideally following from bud formation to flowering, thus including flower or fruit abortion (frost or drought damage of trees about to mast).

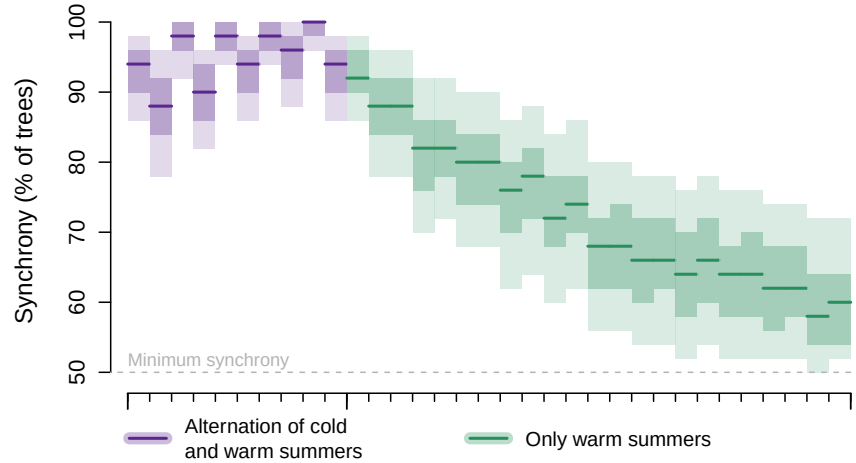


Figure 6: **Cold summers synchronize reproduction at the population level.** We show the predicted percentage of masting trees in a population over time, where the first period includes alternating cold (around 15°C) and warm summers (around 24°C), in purple, and then experiences only warm summers, in green.

Fundamentally synchrony in seed production is one piece of a larger puzzle: understanding future forest regeneration requires disentangling the full sequence of reproductive phenology stages and life stages that lead from a floral bud to an established tree. At the tree-level, new data on bud formation, flowering, and seed production and loss across the multi-year cycle could better forecast masting. Improving our understanding of this first critical step of a much longer sequence could help pinpoint how climate change will ultimately impact forest regeneration. Climate change may lead to failures in the later stages—seed germination, seedling establishment and tree recruitment (Petrie *et al.*, 2023)—even in forests where seed production remains synchronous, making a full life-cycle perspective essential for forecasting regeneration.

Methods

Tree-level reproduction and climate data

We used annual seed production data from 171 individual beech trees (*Fagus sylvatica*) across 12 populations in England, collected between 1980 and 2024, following the protocol described in previous publications (Packham *et al.*, 2008; Bogdziewicz *et al.*, 2021b; Hacket-Pain *et al.*, 2025).

Populations contained between 8 and 21 trees and were monitored for 5 to 45 years. Within populations, individual time series varied in length and contained some missing observations.

We extracted daily climatic data from ERA5-Land (Muñoz-Sabater *et al.*, 2021), at a 0.01 degree resolution, for the 12 sites. We computed three climatic variables, for each year of observation: average maximum temperature in July and August (of the previous year), average spring temperature (April-May), and growing degree days (GDD) until the last frost day as a measure of spring frost risk (Vitasse & Rebetez, 2018).

A Hidden Markov model for individual tree reproduction

We modeled individual tree seed production with a Hidden Markov Model (HMM), in a Bayesian framework using the probabilistic programming language Stan (Carpenter *et al.*, 2017).

Structure of the model

In this model, each tree k is assigned to a latent reproductive state $z_{k,i}$ each year i : a low reproductive state ($z_{k,i} = 1$) or a high reproductive state ($z_{k,i} = 2$). The latent state $z_{k,i}$ is conditionally dependent only on the state the preceding year (the *Markovian* assumption), such that for N years we have:

$$p(z_{k,1}, \dots, z_{k,N}) = p(z_{k,1}) \prod_{i=2}^N p(z_{k,i} \mid z_{k,i-1}) \quad (1)$$

The transitions between reproductive states are determined by a 2×2 matrix of probabilities:

$$\begin{bmatrix} p(z_{k,i} = 1 \mid z_{k,i-1} = 1) & p(z_{k,i} = 2 \mid z_{k,i-1} = 1) \\ p(z_{k,i} = 1 \mid z_{k,i-1} = 2) & p(z_{k,i} = 2 \mid z_{k,i-1} = 2) \end{bmatrix} \quad (2)$$

which is by definition equivalent to:

$$\begin{bmatrix} 1 - p(z_{k,i} = 2 \mid z_{k,i-1} = 1) & p(z_{k,i} = 2 \mid z_{k,i-1} = 1) \\ 1 - p(z_{k,i} = 2 \mid z_{k,i-1} = 2) & p(z_{k,i} = 2 \mid z_{k,i-1} = 2) \end{bmatrix} \quad (3)$$

Both the latent reproductive states and the transition probabilities were inferred from the seed data alone, i.e. the model identify the most probable state sequence for each individual tree given its observed seed counts.

The latent state determines the observational model responsible for the observed seed count $y_{k,i}$. We modeled the seed production in a low reproductive state with a zero-inflated negative binomial distribution, and the one in a high reproductive state with a negative binomial distribution, such as:

$$y_{k,i} \mid z_{k,i} = 1 \sim \begin{cases} 0 & \text{with probability } \theta \\ \text{NegBin}(\lambda_{\text{low}}, \phi_{\text{low}}) & \text{with probability } 1 - \theta \end{cases} \quad (4)$$

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{\text{high}}, \phi_{\text{high}}) \quad (5)$$

Influence of climate on state transition and seed production

We expected the climatic conditions experienced by a tree to influence both the transition matrix (Equation 3) and the seeds produced in a high reproductive state (Equation 5).

In 3, we assumed that only the transition from a low reproductive state to a high reproductive state was affected by climate (and thus, by construction, the probability of staying in a low reproductive as well):

$$p(z_{k,i} = 2 \mid z_{k,i-1} = 1) = \text{logit}^{-1}(\alpha_{1 \rightarrow 2} + \beta_{1 \rightarrow 2}^{\text{summer}} \cdot (X_{k,i-1}^{\text{summer}} - X_{\text{baseline}}^{\text{summer}})) \quad (6)$$

where $X_{k,i-1}^{\text{summer}}$ is the average maximum temperature during summer (the preceding year $i-1$), $\beta_{1 \rightarrow 2}^{\text{summer}}$ is the effect of these summer temperature, and $\text{logit}^{-1}(\alpha_{1 \rightarrow 2})$ is the transition probability when $X_{t,i-1}^{\text{summer}} = X_{\text{baseline}}^{\text{summer}}$.

We did not expect $X_{k,i-1}^{\text{summer}}$ to affect $p(z_{k,i} = 2 \mid z_{k,i-1} = 2)$ because the temporal overlap between fruit development and floral bud initiation should prevent trees already in a high reproductive state to respond to favorable summer conditions (but we still checked that this effect was negligible, see Supp. Fig.).

In 5, we assumed that both spring temperature $X_{k,i}^{\text{spring}}$ and spring frost risk $X_{k,i}^{\text{frost}}$ could impact the seed production in a high reproductive state, such as:

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{k,i}^{\text{high}}, \phi_{\text{high}}) \quad (7)$$

$$\log(\lambda_{k,i}^{\text{high}}) = \log(\lambda_{\text{high}}) + \beta_{\text{high}}^{\text{spring}} \cdot (X_{k,i}^{\text{spring}} - X_{\text{baseline}}^{\text{spring}}) + \beta_{\text{high}}^{\text{frost}} \cdot (X_{k,i}^{\text{frost}} - X_{\text{baseline}}^{\text{frost}}) \quad (8)$$

Masting synchrony and model predictions

Individual tree states were aggregated within and across populations to quantify reproductive synchrony (the proportion of trees sharing the same reproductive state). Synchrony within populations was computed for each population as the maximum of the proportion of trees in a high or low reproductive state, then averaged across populations. Synchrony across populations was computed similarly, but the proportions of trees in each state were first averaged across all populations before taking the maximum (to avoid a population with more trees to have a higher contribution). Synchrony within populations can thus be higher than synchrony across populations when trees within a site tend to be in the same state but differ between sites.

To forecast the effects of summer warming, we simulated seed production (and synchrony) for 50 trees representing an average English population with projected summer warming up to 7°C over one century (Schumacher *et al.*, 2024). To assess the role of biological constraints, we compared predictions from our model—in which reproductive constraints prevent trees to stay indefinitely in a high reproductive state—against a hypothetical model where summer temperatures had a similar effect on both $p(z_{k,i} = 2 \mid z_{k,i-1} = 1)$ (transition from low to high) and $p(z_{k,i} = 2 \mid z_{k,i-1} = 2)$ (persistence in high).

To illustrate the role of cold summers in reproductive synchrony, we also simulated seed production for 50 trees first experiencing a perfect alternation of cold (around 15°C) and warm summers (around 24°C), followed by a period of warm summers only.

1 References

References

- Albrich, K., Rammer, W. & Seidl, R. (2020) Climate change causes critical transitions and irreversible alterations of mountain forests. *Global Change Biology* **26**, 4013–4027.
- Augspurger, C.K. (2009) Spring 2007 warmth and frost: phenology, damage and refoliation in a temperate deciduous forest. *Functional Ecology* **23**, 1031–1039.
- Bangerth, K. (2009) Floral induction in mature, perennial angiosperm fruit trees: Similarities and discrepancies with annual/biennial plants and the involvement of plant hormones. *Scientia Horticulturae* **122**, 153–163.
- Bogdziewicz, M., Hacket-Pain, A., Ascoli, D. & Szymkowiak, J. (2021a) Environmental variation drives continental-scale synchrony of european beech reproduction. *Ecology* **102**.
- Bogdziewicz, M., Hacket-Pain, A., Kelly, D., Thomas, P.A., Lageard, J. & Tanentzap, A.J. (2021b) Climate warming causes mast seeding to break down by reducing sensitivity to weather cues. *Global Change Biology* **27**, 1952–1961.
- Bogdziewicz, M., Kelly, D., Ascoli, D., Caignard, T., Chianucci, F., Crone, E.E., Fleurot, E., Foest, J.J., Gratzner, G., Hagiwara, T., Han, Q., Journé, V., Keurinck, L., Kondrat, K., McClory, R., LaMontagne, J.M., Mundo, I.A., Nussbaumer, A., Oberklammer, I., Ohno, M., Pearse, I.S., Pesendorfer, M.B., Resente, G., Satake, A., Shibata, M., Snell, R.S., Szymkowiak, J., Touzot, L., Zwolak, R., Zywiec, M. & Hacket-Pain, A.J. (2024) Evolutionary ecology of masting: mechanisms, models, and climate change. *Trends in Ecology & Evolution* **39**, 851–862.
- Bogdziewicz, M., Kelly, D., Tanentzap, A.J., Thomas, P., Foest, J., Lageard, J. & Hacket-Pain, A. (2023) Reproductive collapse in european beech results from declining pollination efficiency in large trees. *Global Change Biology* **29**, 4595–4604.
- Bogdziewicz, M., Kelly, D., Thomas, P.A., Lageard, J.G.A. & Hacket-Pain, A. (2020) Climate warming disrupts mast seeding and its fitness benefits in european beech. *Nature Plants* **6**, 88–94.
- Bontrager, M. & Angert, A.L. (2019) Gene flow improves fitness at a range edge under climate change. *Evolution Letters* **3**, 55–68.
- Carlson, C.J., Brookson, C.B., Becker, D.J., Cummings, C.A., Gibb, R., Halliday, F.W., Heckley, A.M., Huang, Z.Y.X., Lavelle, T., Robertson, H., Vicente-Santos, A., Weets, C.M. & Poisot, T. (2025) Pathogens and planetary change. *Nature Reviews Biodiversity* **1**, 32–49.
- Carlson, S.M., Cunningham, C.J. & Westley, P.A. (2014) Evolutionary rescue in a changing world. *Trends in Ecology & Evolution* **29**, 521–530.

- Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P. & Riddell, A. (2017) Stan: A probabilistic programming language. *Journal of Statistical Software* **76**.
- Chakraborty, D., Ciceu, A., Ballian, D., Benito Garzón, M., Bolte, A., Bozic, G., Buchacher, R., Čepl, J., Cremer, E., Ducouso, A., Gaviria, J., George, J.P., Hardtke, A., Ivankovic, M., Klisz, M., Kowalczyk, J., Kremer, A., Lstibůrek, M., Longauer, R., Mihai, G., Nagy, L., Petkova, K., Popov, E., Schirmer, R., Skrøppa, T., Solvin, T.M., Steffenrem, A., Stejskal, J., Stojnic, S., Volmer, K. & Schueler, S. (2024) Assisted tree migration can preserve the european forest carbon sink under climate change. *Nature Climate Change* **14**, 845–852.
- Chamberlain, C.J., Cook, B.I., Morales-Castilla, I. & Wolkovich, E.M. (2021) Climate change reshapes the drivers of false spring risk across european trees. *New Phytologist* **229**, 323–334.
- Daru, B.H., Davies, T.J., Willis, C.G., Meineke, E.K., Ronk, A., Zobel, M., Pärtel, M., Antonelli, A. & Davis, C.C. (2021) Widespread homogenization of plant communities in the anthropocene. *Nature Communications* **12**.
- Davies, T.J. & MacPherson, A. (2024) Seed masting as a mechanism for escape from pathogens. *Current Biology* **34**, R120–R125.
- Dornelas, M., Gotelli, N.J., McGill, B., Shimadzu, H., Moyes, F., Sievers, C. & Magurran, A.E. (2014) Assemblage time series reveal biodiversity change but not systematic loss. *Science* **344**, 296–299.
- Foest, J.J., Bogdziewicz, M., Pesendorfer, M.B., Ascoli, D., Cutini, A., Nussbaumer, A., Verstraeten, A., Beudert, B., Chianucci, F., Mezzavilla, F., Gratzer, G., Kunstler, G., Meesenburg, H., Wagner, M., Mund, M., Cools, N., Vacek, S., Schmidt, W., Vacek, Z. & Hacket-Pain, A. (2024) Widespread breakdown in masting in european beech due to rising summer temperatures. *Global Change Biology* **30**.
- Forzieri, G., Dakos, V., McDowell, N.G., Ramdane, A. & Cescatti, A. (2022) Emerging signals of declining forest resilience under climate change. *Nature* **608**, 534–539.
- Gallego Zamorano, J., Hokkanen, T. & Lehtikainen, A. (2016) Climate-driven synchrony in seed production of masting deciduous and conifer tree species. *Journal of Plant Ecology* p. rtw117.
- Geber, M.A., Watson, M.A. & de Kroon, H. (1997) *Organ Preformation, Development, and Resource Allocation in Perennials*, pp. 113–141. Elsevier.
- Hacket-Pain, A., Szymkowiak, J., Journé, V., Barczyk, M.K., Thomas, P.A., Lageard, J.G.A., Kelly, D. & Bogdziewicz, M. (2025) Growth decline in european beech associated with temperature-driven increase in reproductive allocation. *Proceedings of the National Academy of Sciences* **122**.
- Janzen, D.H. (1971) Seed predation by animals. *Annual Review of Ecology and Systematics* **2**, 465–492.

- Janzen, D.H. (1978) Seeding patterns of tropical trees. *Tropical trees as living systems*, pp. 83–128, Cambridge University Press Cambridge.
- Journé, V., Kelly, D., Hacket-Pain, A., Pearse, I.S., Szymkowiak, J., Foest, J.J., Kondrat, K., Oberklammer, I., Pesendorfer, M.B., Satake, A. & Bogdziewicz, M. (2025) Weather drivers of reproductive variability in perennial plants and their implications for climate change risks. *Nature Communications* **16**.
- Journé, V., Szymkowiak, J., Foest, J., Hacket-Pain, A., Kelly, D. & Bogdziewicz, M. (2024) Summer solstice orchestrates the subcontinental-scale synchrony of mast seeding. *Nature Plants* **10**, 367–373.
- Kelly, D. (1994) The evolutionary ecology of mast seeding. *Trends in Ecology & Evolution* **9**, 465–470.
- Kelly, D., Geldenhuis, A., James, A., Penelope Holland, E., Plank, M.J., Brockie, R.E., Cowan, P.E., Harper, G.A., Lee, W.G., Maitland, M.J., Mark, A.F., Mills, J.A., Wilson, P.R. & Byrom, A.E. (2012) Of mast and mean: differential-temperature cue makes mast seeding insensitive to climate change. *Ecology Letters* **16**, 90–98.
- Kelly, D. & Sork, V.L. (2002) Mast seeding in perennial plants: Why, how, where? *Annual Review of Ecology and Systematics* **33**, 427–447.
- Koenig, W.D. & Knops, J.M.H. (1998) Scale of mast-seeding and tree-ring growth. *Nature* **396**, 225–226.
- LaMontagne, J.M., Pearse, I.S., Greene, D.F. & Koenig, W.D. (2020) Mast seeding patterns are asynchronous at a continental scale. *Nature Plants* **6**, 460–465.
- McDowell, N.G., Allen, C.D., Anderson-Teixeira, K., Aukema, B.H., Bond-Lamberty, B., Chini, L., Clark, J.S., Dietze, M., Grossiord, C., Hanbury-Brown, A., Hurtt, G.C., Jackson, R.B., Johnson, D.J., Kueppers, L., Lichstein, J.W., Ogle, K., Poulter, B., Pugh, T.A.M., Seidl, R., Turner, M.G., Uriarte, M., Walker, A.P. & Xu, C. (2020) Pervasive shifts in forest dynamics in a changing world. *Science* **368**.
- Milyaev, A., Kofler, J., Moya, Y.A.T., Lempe, J., Stefanelli, D., Hanke, M.V., Flachowsky, H., von Wirén, N. & Wünsche, J.N. (2022) Profiling of phytohormones in apple fruit and buds regarding their role as potential regulators of flower bud formation. *Tree Physiology* **42**, 2319–2335.
- Monselise, S. & Goldschmidt, E. (1982) Alternate bearing in fruit trees.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C. & Thépaut, J.N. (2021) Era5-land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* **13**, 4349–4383.

- 388 Nussbaumer, A., Meusburger, K., Schmitt, M., Waldner, P., Gehrig, R., Haeni, M., Rigling, A.,
389 Brunner, I. & Thimonier, A. (2020) Extreme summer heat and drought lead to early fruit abortion
390 in european beech. *Scientific Reports* **10**.
- 391 Nussbaumer, A., Waldner, P., Apuhtin, V., Aytar, F., Benham, S., Bussotti, F., Eichhorn, J.,
392 Eickenscheidt, N., Fabianek, P., Falkenried, L., Leca, S., Lindgren, M., Manzano Serrano, M.J.,
393 Neagu, S., Nevalainen, S., Pajtik, J., Potočić, N., Rautio, P., Sioen, G., Stakėnas, V., Tasdemir,
394 C., Thomsen, I.M., Timmermann, V., Ukonmaanaho, L., Verstraeten, A., Wulff, S. & Gessler,
395 A. (2018) Impact of weather cues and resource dynamics on mast occurrence in the main forest
396 tree species in europe. *Forest Ecology and Management* **429**, 336–350.
- 397 Packham, J.R., Thomas, P.A., Lageard, J.G. & Hilton, G.M. (2008) The english beech masting
398 survey 1980–2007: Variation in the fruiting of the common beech (*fagus sylvatical.*) and its effects
399 on woodland ecosystems. *Arboricultural Journal* **31**, 189–214.
- 400 Petrie, M., Hubbard, R., Bradford, J., Kolb, T., Noel, A., Schlaepfer, D., Bowen, M., Fuller, L. &
401 Moser, W. (2023) Widespread regeneration failure in ponderosa pine forests of the southwestern
402 united states. *Forest Ecology and Management* **545**, 121208.
- 403 Sachser, F., Gratzner, G., Pesendorfer, M.B., Konrad, H., Kempter, I. & Nopp-Mayr, U. (2025)
404 Tracking individual seed fate confirms mainly antagonistic interactions between rodents and
405 european beech. *Biology Letters* **21**.
- 406 Schumacher, D.L., Singh, J., Hauser, M., Fischer, E.M., Wild, M. & Seneviratne, S.I. (2024)
407 Exacerbated summer european warming not captured by climate models neglecting long-term
408 aerosol changes. *Communications Earth & Environment* **5**.
- 409 Trisos, C.H., Merow, C. & Pigot, A.L. (2020) The projected timing of abrupt ecological disruption
410 from climate change. *Nature* **580**, 496–501.
- 411 Vacchiano, G., Hacket-Pain, A., Turco, M., Motta, R., Maringer, J., Conedera, M., Drobyshev,
412 I. & Ascoli, D. (2017) Spatial patterns and broad-scale weather cues of beech mast seeding in
413 europe. *New Phytologist* **215**, 595–608.
- 414 Vitasse, Y. & Rebetez, M. (2018) Unprecedented risk of spring frost damage in switzerland and
415 germany in 2017. *Climatic Change* **149**, 233–246.
- 416 Wernberg, T., Bennett, S., Babcock, R.C., de Bettignies, T., Cure, K., Depczynski, M., Dufois, F.,
417 Fromont, J., Fulton, C.J., Hovey, R.K., Harvey, E.S., Holmes, T.H., Kendrick, G.A., Radford,
418 B., Santana-Garcon, J., Saunders, B.J., Smale, D.A., Thomsen, M.S., Tuckett, C.A., Tuya,
419 F., Vanderklift, M.A. & Wilson, S. (2016) Climate-driven regime shift of a temperate marine
420 ecosystem. *Science* **353**, 169–172.
- 421 Wilkie, J.D., Sedgley, M. & Olesen, T. (2008) Regulation of floral initiation in horticultural trees.
422 *Journal of Experimental Botany* **59**, 3215–3228.

Zohner, C.M., Mo, L.D., Renner, S.S., Svenning, J.C., Vitasse, Y., Benito, B.M., Ordonez, A., Baumgarten, F., Bastin, J.F., Sebald, V., Reich, P.B., Liang, J.J., Nabuurs, G.J., de Miguel, S., Alberti, G., Anton-Fernandez, C., Balazy, R., Brandli, U.B., Chen, H.Y.H., Chisholm, C., Cien-
ciala, E., Dayanandan, S., Fayle, T.M., Frizzera, L., Gianelle, D., Jagodzinski, A.M., Jaroszewicz, B., Jucker, T., Kepfer-Rojas, S., Khan, M.L., Kim, H.S., Korjus, H., Johannsen, V.K., Laar-
mann, D., Lang, M., Zawila-Niedzwiecki, T., Niklaus, P.A., Paquette, A., Pretzsch, H., Saikia, P., Schall, P., Seben, V., Svoboda, M., Tikhonova, E., Viana, H., Zhang, C.Y., Zhao, X.H. & Crowther, T.W. (2020) Late-spring frost risk between 1959 and 2017 decreased in north america but increased in europe and asia. *Proceedings of the National Academy of Sciences of the United States of America* **117**, 12192–12200.